

Abhimanyu Bhola



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Summary

I am an EPFL Masters student studying Microengineering, with more than 2 years of full-time work experience as a mechanical engineer in a Y-combinator backed health-tech startup. I am actively looking for a 4–6 month internship, **as mandated by EPFL for all master's students**, in high-tech product development, microtechnologies, or mechanical hardware engineering during the period of July 2027 to February 2028.

Work Experience

Mechanical Design Engineer – Inito Health Tech Pvt Ltd (YC W19), Bengaluru, India | 

July 2023 – Nov 2025

Patent (Filed) as a co-inventor: System and Method for Imaging a Test Strip

- An **imaging system** used to capture and analyse images from test strips, and provide quantitative hormonal insights

Product Design

- Led end-to-end design and validation of **7 components** in a test strip imaging system, in collaboration with data science and firmware teams, achieving **95% accuracy** in hormone measurements against laboratory benchmarks
- Designed and conducted 20+ experiments during the development cycle to test and validate multiple designs, using simulations, **SLA 3D printing** for prototyping and scaling up to **injection moulding** for production
- Improved reliability and precision of the device, **achieving detection of 2× fainter signals** compared to the older generation; incorporated a UV LED to support Europium-based assays
- Performed a detailed **Design Failure Modes and Effects Analysis (DFMEA)** on the imaging system, singling out key mechanical failure modes in the system; made design changes for the highest risk failure modes

Production, Quality and Vendor Management

- Developed assembly and quality control (QC) protocols for a pilot build of 6000 devices, at a scale of **300 devices per week**; trained the quality and production engineers in these protocols
- Designed the **imaging test target for the end-of-line QC** by experimenting with screen, pad and digital printing
- Conducted on-site evaluations at 4 vendors across China to validate production of critical plastic and optical components including **cameras, first-surface mirrors, and beamsplitters**

Battery Engineering Intern – Ather Energy, Bengaluru, India | 

Dec 2021 – May 2022

- Researched and experimented on laser welding of cell to busbar linkages in a battery pack for electric two-wheelers
- Developed Python scripts to characterize and measure battery pack imbalances of Ather battery packs in the field, and performed an RCA of >15 imbalance battery packs by working with the cell engineering team

Education

École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland

Sep 2026- Present

MS, Microengineering

- Awarded the [K.C. Mahindra Education Trust Postgraduate Scholarship](#) 2026 to help fund my education
- **Relevant Coursework:** Micro/Nanomechanical Devices, Introduction to Additive Manufacturing, Product Design and Systems Engineering, Fundamentals of Biomicroscopy, Image Processing

Indian Institute of Technology Madras (IITM), Chennai, India

Jul 2018- Jul 2023

B. Tech in Engineering Design & M. Tech in Automotive Engineering (Dual Degree Program) [GPA: 9.01/10]

- **Relevant Coursework:** Mechatronics System Design, Design of Mechanical Systems, Physics of Measurement, Control Systems, Vehicle Dynamics, Smart Sensing for IoT, Pattern Recognition and Machine Learning

Research Experience

Graduate Researcher – Virtual Proving Ground and Simulations Lab, IITM |

Jun 2022 – Jul 2023

Master's Thesis: Reinforcement Learning for Battery Charging Algorithms

- Developed a model of an LFP cell on MATLAB Simulink to simulate its electrical, thermal and aging characteristics
- Designed a Proximal Policy Optimisation based RL algorithm in **MATLAB** for the fast charging of the cell, while ensuring **fast charging in an ambient temperature range of 10°C to 40°C**
- Achieved a **50% reduction in charging time** at the same ambient temperature when compared to 1C CCCV charging, while maintaining a similar battery degradation
- The abstract was selected for the Cyber Physical Systems Summit 2023, held at IIT Kharagpur

Undergraduate Researcher – Young Research Fellow Program, IITM

Sep 2020 – Jun 2021

Research Project: Obstacle Avoidance and Target Driven Guidance Algorithm

- Developed a Target Driven Motion Planning and Obstacle Avoidance algorithm for Autonomous Aerial Vehicles in 2D fields using Proportional Navigation and Acceleration Velocity Obstacles
- Achieved a 20% faster and more computer efficient algorithm compared to classical aerospace algorithms

Projects

Team Anveshak (Mars Rover Team) – Mechanical Engineer, Centre for Innovation, IITM |

Feb 2019 – May 2020

- Developed a lightweight, carbon-fibre based rocker-bogie suspension to minimize chassis vibration
- Designed and fabricated custom 3D printed TPU tyres for the wheels, achieving a **weight reduction of 10%**
- Designed a modular, low-cost onboard science laboratory for collection and analysis of soil samples for biomarkers
- Achieved 5th position in the Indian Rover Challenge 2020, qualified for the University Rover Challenge 2020 finals

Anti-Lock Brake System Modelling – Course Project: Vehicle Dynamics

Jan 2021 – May 2021

- Developed a State-Space model based on vehicle and tyre parameters in MATLAB and a bicycle model in Simulink
- Designed PID and Bang-Bang controller for **Anti-lock Braking System (ABS)** for the bicycle model
- Evaluated yaw rate response and lateral acceleration response with Mimuro Plot with varying tyre parameters

IoT Based Interactive Game – Course Project: Smart Sensing for IoT

Jul 2021 – Nov 2021

- Developed a virtual tennis game for Windows using Unity, which used an Android phone as a tennis racket by utilising the IMU sensor data
- Developed a random forest model to classify gestures into four different types of shots as input for the game

Technical Skills

Software and Programming Languages: SolidWorks, Fusion 360, MATLAB, Simulink, Python, C++, Mathematica

Skills: Mechanical design, Rapid prototyping, Design for Manufacturability (DFM) & Assembly (DFA), Knowledge of Injection Moulding & Compression Moulding, Root Cause Analysis, Design of Experiments

College Associations and Volunteer Work

Volunteer Math Teacher – Shikshantar, Mumbai, India

Jan 2026 – Aug 2026

- Volunteered as a Math teacher for Grade 8 and 9 students from the underprivileged communities of Mumbai

Department Legislator and Placement Coordinator, Dept. of Engineering Design, IITM

Jul 2020 – May 2021

- Acted as a representative for the student body of the Engineering Design department
- Resolved institute-wide student concerns as a legislator in the Student Legislative Council, including establishing a COVID-19 grievance committee

Miscellaneous

Language Proficiency: English: C2 (Fluent / Native-level)

Interests: Cooking, Lego, Motorcycling, Running, Gaming, Football, Consumer Electronics